

AI Employee OS — Product Overview & Vision

Product Requirements Document (PRD)

1. Vision

AI Employee OS aims to be the unified operating system for modern small-to-medium businesses (SMBs)

By combining essential tools (CRM, invoicing, tasks, HR) with intelligent "AI Employees", the platform eliminates the need for fragmented software stacks and allows companies to scale their operations without scaling their headcount proportionally.

2. Goals

- **Consolidation:** Replace separate CRM, HR, and accounting tools with a single unified platform.
- **Automation:** Introduce AI agents capable of performing repetitive tasks autonomously.
- **Scalability:** Provide a multi-tenant architecture that securely supports thousands of organizations on a single deployment.

3. Business Objectives

- Reduce software licensing costs for target SMBs by 50%.
- Increase team productivity by offloading routine tasks to AI Employees.
- Provide a robust API and webhook system for enterprise integration.

4. Personas

- **Platform Super Admin:** Manages global SaaS operations, subscriptions, and module enablement.
- **Company Admin (Org Admin):** The business owner who configures the platform, adds employees, and oversees company performance.
- **Employee:** A staff member interacting with role-specific dashboards (e.g., Sales Manager, HR).
- **AI Employee:** Autonomous agents executing tasks like drafting emails, categorizing expenses, and updating CRM statuses.

5. Functional Requirements

5.1 Authentication & Tenancy

- Users must authenticate via Supabase Auth (Email/Password or OAuth).
- All data must be strictly isolated by `organization_id` using Postgres Row Level Security (RLS).
- Users cannot access data across tenants unless designated as a Super Admin.

5.2 Business Modules

- **CRM:** Manage customers, leads, pipelines, and activities.
- **Finance:** Issue quotations and invoices, track payments.
- **HR:** Manage employees, attendance, and leave requests.
- **Documents:** Upload, store, and search company documents (Knowledge Base).

5.3 AI Capabilities

- Provide interactive chat interfaces to communicate with specific AI agents.
- Agents must have access to internal tools (e.g., "Search CRM", "Generate Invoice").

- The system must maintain long-term memory using pgvector-based RAG.

6. Non-Functional Requirements

- **Performance:** UI rendering must be sub-500ms. AI responses must stream instantly.
- **Security:** RLS must be enforced at the database level. API keys and secrets must be encrypted.
- **Scalability:** The backend must handle high-throughput async requests via FastAPI and Celery.

7. User Stories

- *As a Company Admin, I want to invite employees to my organization so they can access their dashboards.*
- *As a Sales Rep, I want to ask the AI Sales Assistant to summarize a lead's recent activity.*
- *As a Super Admin, I want to view platform-wide billing and usage statistics.*

8. Acceptance Criteria (MVP)

- A user can register, creating an organization automatically.
- The user can navigate to the CRM and manually add a lead.
- The user can open the AI Chat interface and receive a text response from an LLM.
- Database queries across different organizations yield zero results (tenant isolation verified).

9. Future Roadmap

- Voice-enabled AI employees.
- Deep integration with Slack and Microsoft Teams.
- Advanced multi-agent orchestration (agents delegating tasks to other agents).